TEAM MEMBERS

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PROJECT PROPOSAL

APEXUCL: Autonomous 36-Team Swiss-Model League & Knockout Orchestration System for the UEFA Champions League

Project Overview

ApexUCL is a desktop-based tournament governance and scheduling system engineered specifically for the 36-team Swiss-model format of the UEFA Champions League. Replacing the legacy four-team group stage, this modern format consolidates all 36 clubs into a single unified league table where each club plays eight distinct matches (four home, four away) against opponents drawn systematically across four seeding pots.

The system automates complex constraint-driven fixture scheduling, computes real-time standings using the official UEFA multi-tier tie-breaker hierarchy, and coordinates the structural transition from the league phase into the knockout phase. Through a desktop GUI, administrators can configure pot seedings, record matchday scorelines, update standings dynamically, and generate both the two-legged knockout play-off pairings and the subsequent Round of 16 bracket.

Motivation

The expansion to a 36-team single-table tournament introduces severe combinatorial complexity that makes manual scheduling and spreadsheet-based operations prone to errors. Generating the 144 league phase fixtures requires satisfying strict operational constraints simultaneously: ensuring balanced home and away distribution across pots, enforcing country protection rules (preventing domestic league clashes), and capping foreign association exposure (no more than two opponents from any single external league).

Furthermore, determining rankings in a unified 36-team table requires calculating complex multi-tier tie-breakers (Goal Difference, Goals Scored, Away Goals, Total Wins, and UEFA Club Coefficients) where boundary margins determine direct qualification, play-off relegation, or outright elimination. ApexUCL addresses these challenges by translating intricate tournament regulations into robust Software Design and Analysis (SDA) principles, delivering an autonomous, verifiable, and rule-compliant tournament management engine.

Objectives

- ***Constraint-Satisfaction Fixture Engine***: Design and implement a scheduling engine to generate all 144 league phase matches ensuring exact pot distribution (two opponents from each of Pots 1 through 4), strict 4-home/4-away balance, and country-protection boundaries.
- ***Autonomous Standings & Tie-Breaker Architecture***: Build an extensible table calculation engine using the Strategy Design Pattern to dynamically evaluate and sort the 36-team table according to UEFA's official multi-tier tie-breaker hierarchy.

- ***Automated Knockout Progression Mapping***: Model and automate the tripartite qualification logic that partitions the final league table into Direct Round of 16 qualifiers (Ranks 1-8), Knockout Play-off contenders (Ranks 9-24), and Eliminated clubs (Ranks 25-36).
- ***Two-Legged Play-off Governance***: Manage aggregate scorelines, away match logistics, and extra-time/penalty shootouts for the play-off phase to seed winners into the fixed-path Round of 16 bracket.
- ***Desktop Governance Console***: Provide a clean, structured desktop graphical user interface enabling tournament directors and officials to view tables, record match events, and inspect interactive knockout trees.

Expected Outcomes

- ***Verified Fixture Matrix***: A fully validated 144-match tournament schedule completely devoid of domestic league clashes and compliant with UEFA pot-allocation policies.
- ***Dynamic 36-Team League Dashboard***: A live table visualizer that color-codes the three qualification zones (Ranks 1-8 Green, Ranks 9-24 Blue, Ranks 25-36 Red) and updates standings instantaneously upon match result submission.
- ***Algorithmic Knockout Bracket Generator***: An automated bracket engine that pairs seeded teams (Ranks 9-16) with unseeded teams (Ranks 17-24) and routes aggregate winners directly to their predetermined Round of 16 paths.
- ***Persistent Tournament Data Store***: A fully normalized relational database schema preserving club coefficients, historical match events, disciplinary records, and stage transitions.
- ***Formal SDA Design Artifacts:*** A software design portfolio including UML Use Case models, Domain Class Diagrams, Sequence Diagrams, and an extensive Requirements Traceability Matrix (RTM).

Project Scope

In Scope:

- Club registration, coefficient ranking, and automated seeding into four 9-team pots.
- Backtracking-based constraint fixture generation for the 36-team Swiss model.
- Match event recording (scores, home/away status, matchday completion state).
- Implementation of the complete UEFA tie-breaker algorithm using the Strategy Design Pattern.
- Standings state management and table partitioning (R16 bye, Play-offs, Elimination).
- Two-legged Knockout Play-off fixture pairing, aggregate scoring evaluation, and Round of 16 bracket initialization.
- Local relational database management for transactional consistency and integrity.

Out of Scope:

- Public web or mobile live-streaming and ticketing sales infrastructure.
- Player physical telemetry, biometrics, or GPS tracking integrations.
- Direct financial distribution, media rights, or club commercial budgeting models.
- Manual referee assignment logistics or VAR video streaming integrations.

Tools and Technologies

- ***Frontend (GUI)***: Java (JDK 17+) using JavaFX (or Java Swing) for a structured, modular desktop interface (standings dashboards, fixture matrix grids, and interactive knockout tree components).
- ***Core Logic & Architecture:*** Java Object-Oriented paradigm utilizing core GoF Design Patterns (Strategy Pattern for tie-breakers, Observer Pattern for match-to-standings updates, Factory Pattern for knockout bracket generation).
- ***Backend & Database:*** MySQL / Oracle Database / SQL Server for persistent relational storage, utilizing JDBC for transaction processing and schema integrity (Clubs, Pots, Fixtures, MatchResults, StandingsEntry, KnockoutPairings).
- ***Design & Analysis Tools:*** Visual Paradigm / Enterprise Architect / Draw.io for standard UML modeling (Use Case Diagrams, Class Diagrams, Sequence Diagrams, and Activity Diagrams).
- ***Testing & Quality Assurance:*** JUnit 5 for unit testing and Boundary Value Analysis (qualification margins: 8th vs. 9th rank and 24th vs. 25th rank).